Experiment No. – 1

Aim : To configure a **router** by connecting a **PC** to it using console access and setting **basic router parameters**

Physical Topology :

Commands used :

1. **enable-** Enters privileged EXEC mode.
2. **configure terminal** – Enters global configuration mode.
3. **hostname Kshitij2300320100131**– Assigns the name Kshitij2300320100131 to the switch for identification.
4. **line console 0** – Accesses the console line configuration mode.
5. **password helloworld** – Sets the console line password as *helloworld*.
6. **login** – Enables password checking on the console line.
7. **exit** – Exits the current configuration mode.
8. **enable secret class** – Sets an encrypted privileged (enable) password.
9. **line vty 0 15**– Accesses virtual terminal lines for remote access.
10. **password helloworld** – Sets the VTY line password as *helloworld*.
11. **login** – Enables password authentication on VTY lines.
12. **banner motd #Unauthorized access is prohibited#** – Displays a warning message to users when they access the router**.**

STEP-1:renaming the networking device switch

```
Kshitij 2300320100131

Kshitij>enable
Password:
Kshitij#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Kshitij(config)#hostname Kshitij2300320100131
Kshitij2300320100131(config)#line console 0
```

STEP-2: Setting password on console line ,virtual teletype and priviledge level, also including message of the day

```
Kshitij 2300320100131

Kshitij>enable
Password:
Kshitij#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Kshitij(config)#hostname Kshitij2300320100131
Kshitij2300320100131(config)#line console 0
Kshitij2300320100131(config-line)#password helloworld
Kshitij2300320100131(config-line)#login
Kshitij2300320100131(config-line)#exit
Kshitij2300320100131(config)#enable secret class
Kshitij2300320100131(config)#line vty 0 15
Kshitij2300320100131(config-line)#password helloworld
Kshitij2300320100131(config-line)#login
Kshitij2300320100131(config-line)#exit
Kshitij2300320100131(config)#service password-encryption
Kshitij2300320100131(config)#banner motd #Welcome to router configuration!#
Kshitij2300320100131(config)#exit
Kshitij2300320100131#
%SYS-5-CONFIG_I: Configured from console by console

Kshitij2300320100131#write memory
Building configuration...
[OK]
Kshitij2300320100131#
```

Visuals of running-config: after encryption

```
!
line con 0
 password 7 082949420516121800070 8
 login
!
line aux 0
!
line vty 0 4
 password 7 082949420516121800070 8
 login
line vty 5 15
 password 7 082949420516121800070 8
 login
!
```

STEP-3: Accessing configuration of device and its level by using password

```
Welcome to router configuration!

User Access Verification

Password:

Kshitij2300320100131>enable
Password:
Kshitij2300320100131#show running-config
Building configuration...

Current configuration : 929 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Kshitij2300320100131
'
```